DATA ANALYSIS

ZOMATO

Optimizing Rider Assignment & Customer ETA through Signal-Level Improvements

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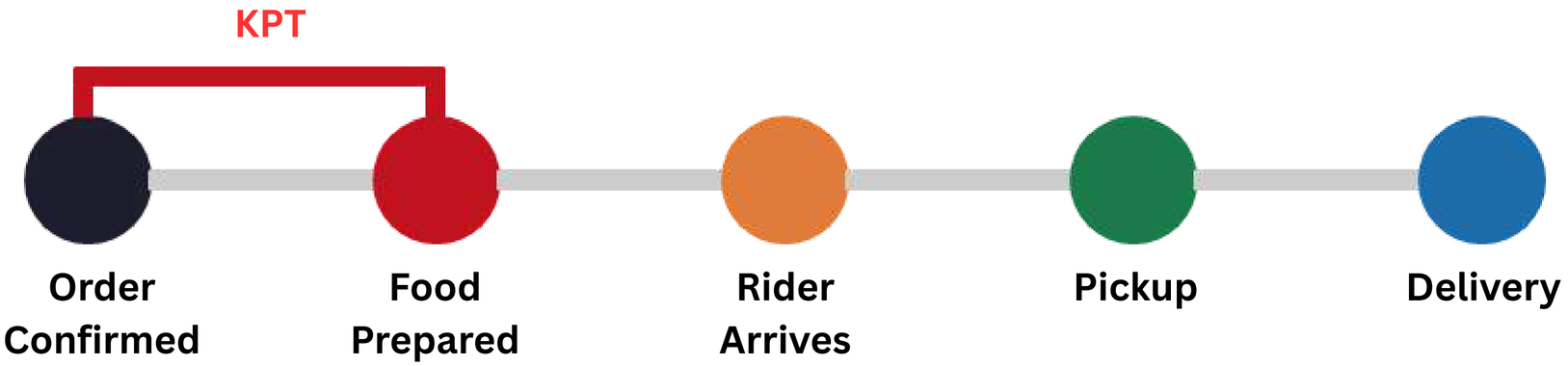
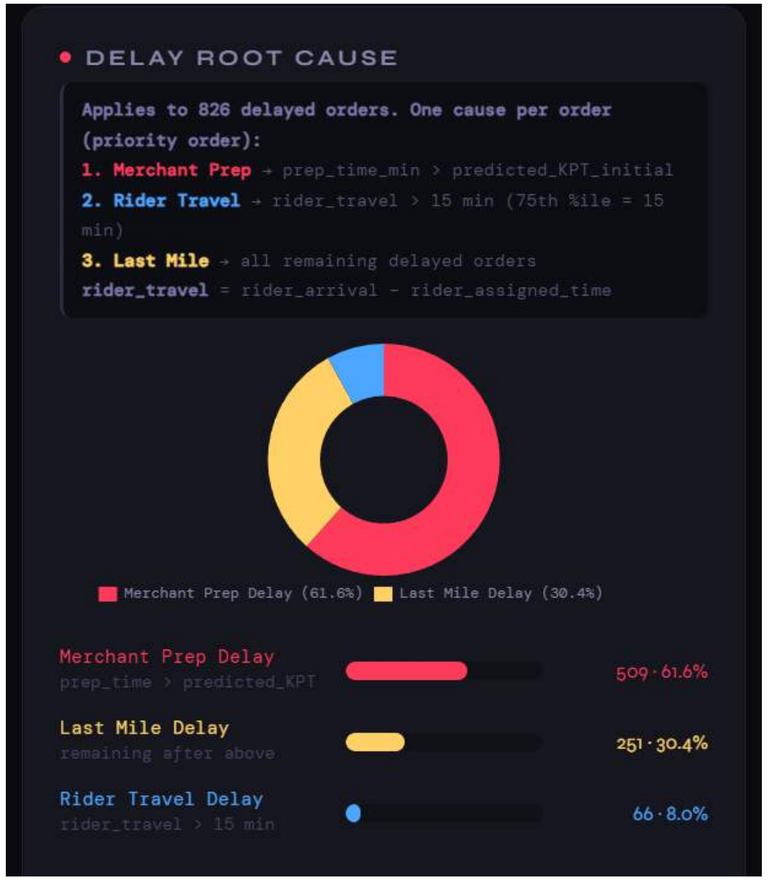
Problem Statement

How might we improve the accuracy of Kitchen Prep Time (KPT) predictions by enriching system-level input signals beyond manual merchant clicks, while ensuring the solution scales across hundreds of thousands of merchants?

What is Kitchen Prep Time (KPT)?

KPT = Time from order confirmation to food being ready for pickup





Why KPT Matter?

Accurate ETA	Rider Efficiency	Lower Costs	300K+ Merchants
Customers get reliable delivery time promises	Riders arrive exactly when food is ready	Zero idle time at restaurant = cost savings	Even 1-min improvement = massive scale impact

Pain Points– Why Current KPT Signals Fail

PAIN 1

Rider-Influenced FOR Marking — **75.1% of orders (936 orders)**

Merchants mark food ready when the rider physically arrives — not when food is actually prepared. This teaches the KPT model an inflated, biased label.

PAIN 2

No Cross-Platform Visibility — **Invisible load (Unquantified)**

Zomato sees only its own orders. Swiggy orders, dine-in customers, takeaway walk-ins all compete for the same kitchen. Hidden congestion makes KPT unpredictable.

PAIN 3

Human & Operational Bias — **Prep std dev ±12.33 min**

Marking behavior varies by merchant, staff shift, and time of day. Some bulk-mark orders. Some forget to mark during rush. This creates systematic noise.

PAIN 4

Equal Treatment of All Merchants — **82.7% delay rate (low reliability)**

The current model applies identical KPT prediction logic to all 300K merchants. Unreliable merchants get the same buffer as consistent ones — over-trusting bad actors.

Solution 1

Rider Wait Feedback Loop

The Core Issue:

merchant_marked_FOR_time is used as the KPT training label.
But merchants mark LATE (S1) or mark EARLY and food waits (S2).
Both introduce systematic bias into the model.

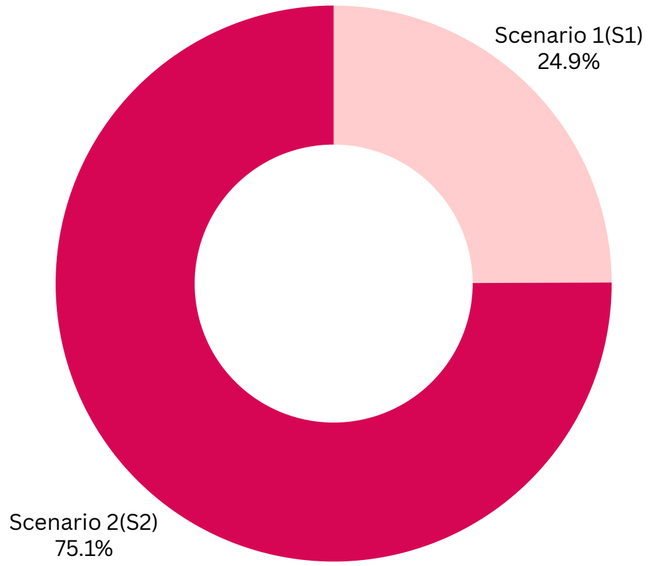
Solution:

Uses rider wait time as an independent observable signal to correct labels.

Correction Logic:

IF S1 (merchant marked Too early):
Corrected KPT = rider_arrival – order_confirmed
→ rider arrival = true food ready signal
Logic: Rider's arrival becomes the true "food ready" signal.

IF S2 (FOR mark time ≈ Rider arrival time):
Formula: Corrected KPT = FOR marked - order_confirmed - 1
Iterative Penalty: Subtracts 1 min continuously to pull dispatch forward and eliminate food idle time.
Convergence: Iteration halts exactly when S2 becomes S1. If KPT is 1 min too short (rider waits 1 min), it adds +1 back, locking in the perfect prediction.



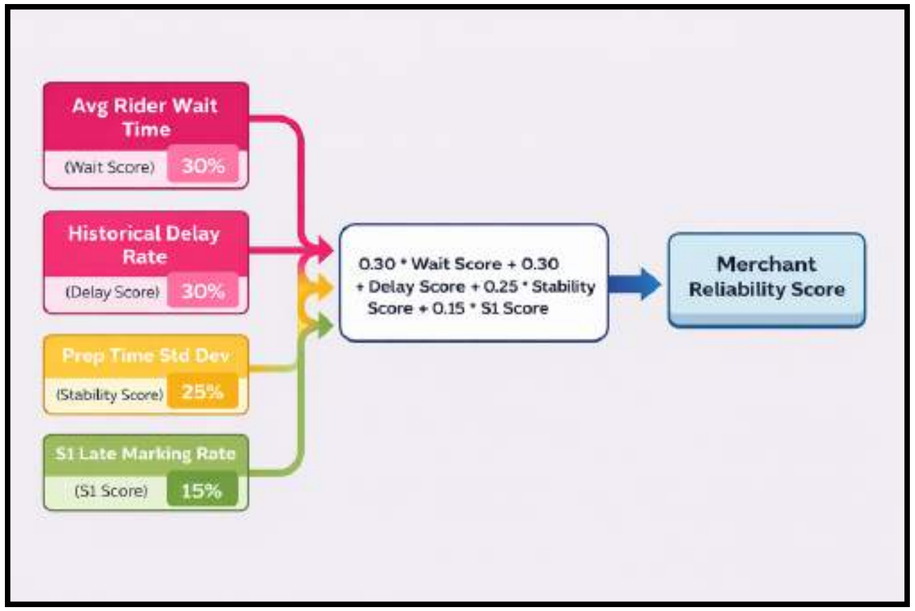
Complete Order data can be segemnted into 2 group
Group 1 FOR marked by merchant too early
Group2 FOR mark time ≈ Rider arrival time

Solution 2

Merchant Reliability Scoring

The Core Issue:

The current system applies identical prediction logic to all 300,000 merchants, regardless of their behavioral history. The system should not trust all merchants equally.



Scoring Logic: A dynamic 0-1 Reliability Score is computed using four signals

30%	+	30%	+	25%	+	15%
Avg Rider Wait Time (Lower wait = more reliable)		Historical Delay Rate (Fewer delays = more reliable)		Prep Time Std Dev (Lower variance = more consistent		S1 Late Marking Rate (Less S1 = accurate marking).

Cold Start Handling (New Merchants)

- Prior: Use cuisine_type + kitchen_type average as starting reliability
- Cloud kitchens: Start at score 0.55 (OTD 37.5% > Dine-in 30.8% in our data)
- Update: After 10+ orders, compute actual score and replace prior

Solution 3

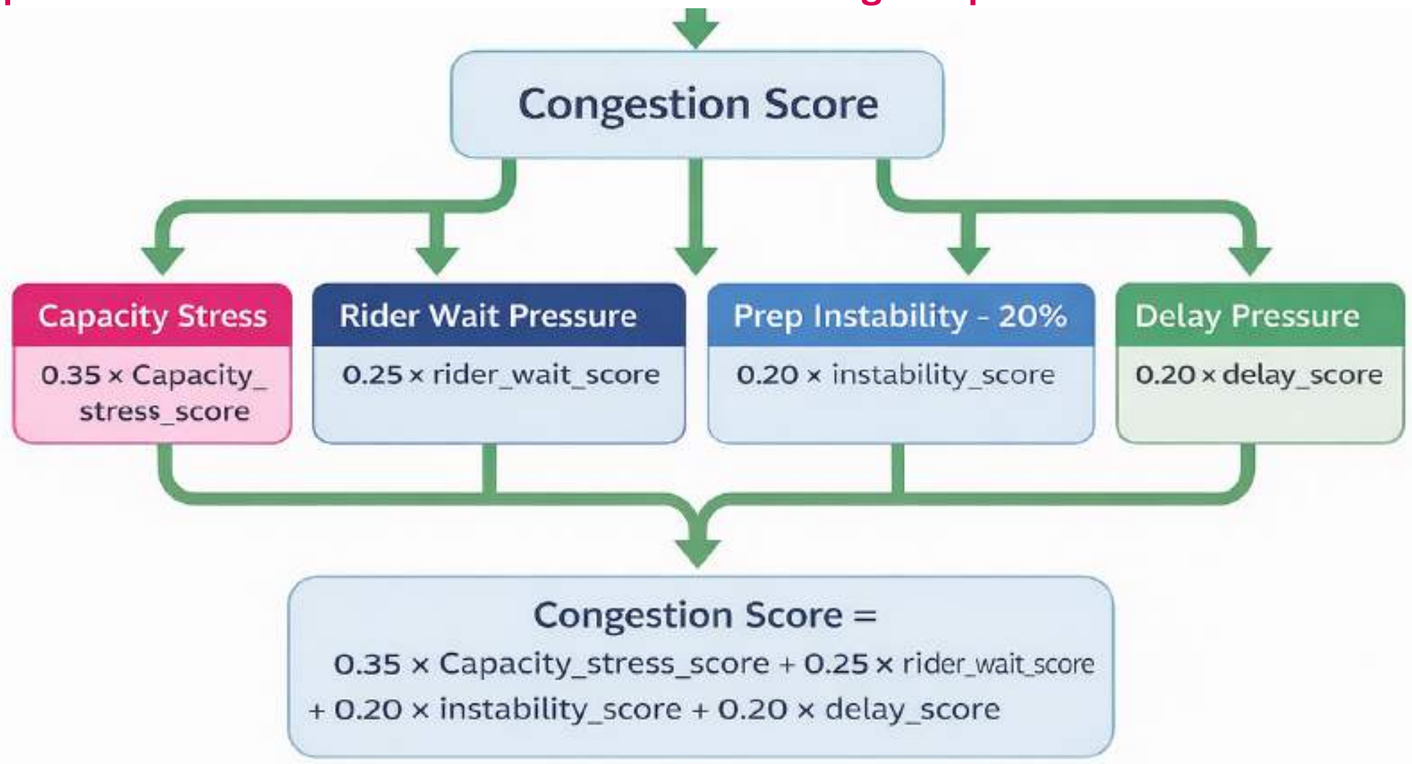
Multi-Signal Kitchen Congestion Estimator

The Core Issue:

The current KPT model has zero visibility into hidden kitchen congestion. It only observes Zomato orders, missing the simultaneous load from Swiggy, dine-in, and takeaways. Active Zomato orders alone have a weak -0.017 correlation with KPT error.

Solution:

Calculates a dynamic Congestion Score (0 to 1) using external proxy signals to capture the true kitchen-wide rush without needing competitor data.



1 Capacity Stress — 35%

- Direct measure of operational pressure
- Captures baseline overload
- Most stable predictor of congestion

2 Rider Wait Pressure — 25%

- Ground-truth operational friction
- Directly affects ETA and rider idle cost
- Strong but slightly reactive signal

3 Prep Instability — 20%

- Captures hidden operational instability
- Indicates surge sensitivity
- Secondary signal compared to structural load

4 Delay Pressure — 20%

- Reflects chronic congestion tendency
- Important for long-term strain assessment
- Complementary to real-time signals

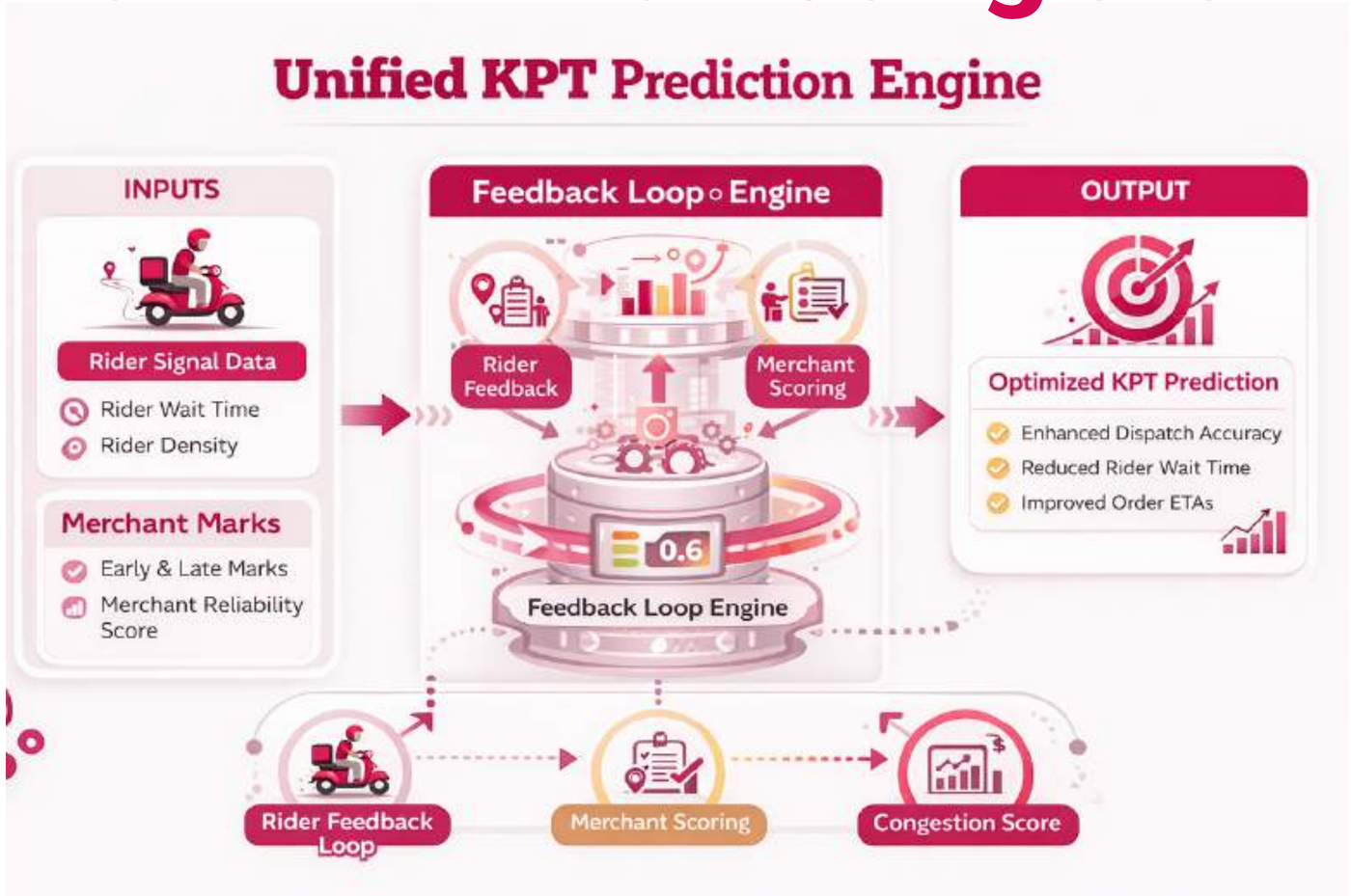
```
Congestion Score =
  0.35 × Capacity_stress_score
+ 0.25 × rider_wait_score
+ 0.20 × instability_score
+ 0.20 × delay_score
```

Final Optimized Solution For Estimating the Corrected KPT

1. Signal Correction (The Foundation):
Cleans raw input data using the Rider Wait Feedback Loop.
Overrides manual "Late Marking" and iteratively penalizes "Early Marking" to eliminate unmeasured idle time.

2. Environmental Overlay (The Context):
Applies a Congestion Proxy Score based on rider density and prep time variance (15.80 to 17.55).
Adjusts for hidden kitchen load that Zomato cannot see directly.

3. Behavioural Adjustment :
1) Increase if congestion is high
2) Increase if reliability is low
3) Increase if riders are actually waitimg



Congestion Adjustment:
Base KPT×C

If congestion = 0.5
→ Add 50% extra prep time.

Reliability Adjustment:
Base KPT×(1−R)

If reliability = 0.8
→ Add 20% risk buffer.

If reliability = 0.3
→ Add 70% risk buffer.

Adjusted KPT=Base KPT+(Base KPT×C)+(Base KPT×(1−R))+(Avg Rider Wait)

Implementation - Ground level Roll Out Plan

PHASE 1 — Week 1-4

Foundation — Sol 1

1. Extract rider_wait signal from existing pickup logs
2. Build S1/S2 classifier (merchant_marked vs rider_arrival)
3. Compute corrected KPT labels for last 90 days
4. Validate: compare corrected vs raw label distribution
5. Shadow mode: run corrected labels alongside existing without changing dispatch

PHASE 2 — Week 5-10

Enrichment — Sol 2 + Sol 3

1. Build congestion score using rider density + variance spike
2. Compute per-merchant reliability score (0-1)
3. Score all 100K+ active merchants using 90-day history
4. Cold start: assign cuisine+kitchen_type peer average
5. A/B test: apply adjusted KPT buffers on 10% of merchants

PHASE 3 — Week 11-16

Monitoring and Scaling

1. Regular KPI monitoring
2. Full rollout after A/B test shows improvement in KPIs
3. Monitor S1 rate daily — target: reduce from 20% to <8%
4. Reliability scores auto-update on rolling 30-day window
5. OTD must Improve by 10-12%

BEFORE vs AFTER – KPI Impact Measurement

KPI Metric	BEFORE	TARGET AFTER	How
Avg Rider Wait at Pickup	5 min	<3min	Sol 1 corrects premature dispatch
ETA Prediction Error (P50,P90)	(21min,45min)	(10,21)	Sol 1+2+3
Order Delay Rate	66.24%	<50%	Sol 1 + 3 preventing early marking + reliability buffers
Rider Wait time Delay Rate	10.34%	1-2%	Sol 1 by reducing noise in KPT
Order Cancellation Rate	10.8%	7-8%	Sol 1+2+3

OTD Dashboard

ZOMATO · MERCHANT OPERATIONS · MARCH 2024

Source: zomato_orders_sample.xlsx + merchant_performance.xlsx
Period: 01–30 Mar 2024 · **Total Orders:** 1,247 · **Merchants:** 100
Peaks: Lunch 12–3 PM · Dinner 7–11 PM

OTD RATE	DELAYED	CANCEL RATE	AVG PREP	AVG RIDER TRAVEL	AVG LAST MILE	AVG DELAY AMT
33.8%	826	10.8%	38.4m	12.4m	29.5m	23.3m
421 on-time	66.2% of total	135 orders	all orders	to reach merchant	pickup→delivery	delayed orders only
COUNT(delayed_flag=0) / 1247 × 100	delivery_time > predicted_delivery_time	SUM(cancellation_flag) / 1247 × 100	MEAN(merchant_marked − order_confirmed)	MEAN(rider_arrival − rider_assigned)	MEAN(delivery_time − pickup_time)	MEAN(delivery − predicted_delivery | flag=1)

OVERALL OTD

OTD% = COUNT(delayed_flag = 0) / COUNT(order_id) × 100
= 421 / 1247 × 100 = **33.8%**
delayed_flag = 1 if delivery_time > predicted_delivery_time else 0

33.8%
ON TIME DELIVERY RATE

421 ON TIME	826 DELAYED
34.6% DINNER OTD	32.6% LUNCH OTD

OTD % DAILY TREND — MARCH 2024

Per day: COUNT(delayed_flag=0 AND date=D) / COUNT(date=D) × 100 · Dashed line = overall avg (33.8%) · Hover for daily order count

DELAY ROOT CAUSE

Applies to 826 delayed orders. One cause per order (priority order):

1. **Merchant Prep** → prep_time_min > predicted_KPT_initial
2. **Rider Travel** → rider_travel > 15 min (75th %ile = 15 min)
3. **Last Mile** → all remaining delayed orders

rider_travel = rider_arrival − rider_assigned_time

Merchant Prep Delay prep_time > predicted_KPT		509 · 61.6%
Last Mile Delay remaining after above		251 · 30.4%
Rider Travel Delay rider_travel > 15 min		66 · 8.0%

Appendix 1

Data Deep-Dive & Label Integrity

Dataset Overview

Sample Size: **1,247** completed orders
Merchants Covered: Multi-category (cloud + dine-in hybrid kitchens)
Time Coverage: Peak + Non-peak windows

Operational Label Integrity Findings

Scenario Split (Ground Truth Analysis):
75.1% of merchants (Scenario 2) mark Food Ready (FOR) after rider arrival
Only **24.9%** (Scenario 1) exhibit premature marking behavior

Statistical Backing

Correlation (Merchant Reliability vs Rider Wait): **0.532**
Indicates strong behavioral signal
Justifies inclusion of reliability as a weighted adjustment layer

Behavioral Adjustment Layer

Final Reliability Score = Weighted composite of 4 merchant behavior signals.

Metric	Weight	Why This Weight?
Avg Rider Wait	30%	Direct financial drain; strongest empirical correlation to KPT error
Historical Delay Rate	30%	Primary contributor to OTD failure & churn risk
Prep Time Variance	25%	Consistency proxy; variance jumps from 15.80 → 17.55 during rush
Late Mark Rate (S1)	15%	Measures process hygiene; weaker than operational delay

Appendix 2

Data Deep-Dive & Label Integrity

Assumptions

1. Reliability Score Categorization

Reliability scores are grouped into four categories:

- High Reliability: Score ≥ 0.75
- Moderate Reliability: 0.60 – 0.74
- Risky: 0.45 – 0.59
- Low Reliability: Score < 0.45

2. Congestion Score Categorization

Congestion levels are defined as follows:

- Severe Congestion: Score ≥ 0.75
- High Congestion: 0.55 – 0.74
- Moderate Congestion: 0.35 – 0.54
- Low Congestion: Score < 0.35

Behavioral Adjustment Layer

Final Congestion score = Weighted composite of 4 proxy signals.

Metric	Weight	Why This Weight?
Capacity Stress Ratio	35%	Direct measure of operational pressure
Rider Wait for Food	25%	Directly affects ETA and rider idle cost
Prep Time Instability	20%	Secondary signal compared to structural load
Delay Pressure	20%	Reflects chronic congestion tendency

Dataset Links

Zomato Orders Dataset

https://docs.google.com/spreadsheets/d/1qC1GQ606XoGxHjob17DloTxdiXXABtP3/edit?usp=sharing&ouid=104015938540875365368&rtpof=true&sd=true

Rider Activity Dataset

https://docs.google.com/spreadsheets/d/1Q-6CgcgKVVziZnz7iRQ9EJBV9NTeseOV/edit?usp=sharing&ouid=104015938540875365368&rtpof=true&sd=true

Merchant Performance Dataset

https://docs.google.com/spreadsheets/d/1sbhNaCrCOxxvmu9clEUtR1LVlmFeqmcT/edit?usp=sharing&ouid=104015938540875365368&rtpof=true&sd=true

Google Colab Link

Thank you!